

Institute of Actuaries of India

Subject ST8 – General Insurance: Pricing

September 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: The Balance sheet of a general company, like any other company, is a summary of financial status of the company at a given moment.

The balance sheet is divided into two parts; Assets and Liabilities.

General Insurance Company's primary assets are Investments, Fixed Assets and Current Assets. Depending on the prevailing regulations, current assets can be Net of Current liabilities or can be gross assets, in which case current liabilities shall be shown on liability side.

The primary liabilities are Technical Reserves, Free Assets and current liabilities (depending on whether current assets are gross or net)

Fixed Assets: Office building, furniture, computers etc.

Investments: Bonds, Equity, property etc.

Current Assets: money due from brokers etc.

Technical reserves: Money set aside in respect of expected claim payments to or on behalf on policyholders, plus related expenses.

Free Reserves: The balancing item equal to the excess of assets over insurance liabilities.

[3 Marks]

Solution 2:

- i) Pin code shall have a large number of levels. A model having this as a factor will have a great deal of random noise and lack of credibility.

Spatial smoothing allows the model the model to fit many values to the pin code factor and then removes the noise from the predictions by allowing the relativity for each pin code to take account of neighbouring values [1]

ii)

- Adjacency based method seems to be more suitable.
- The location based codes tends to be smaller in urban areas and larger in rural areas. It is very rare for a very urban post code to be adjacent to a very rural postcode, so adjacency based smoothing handle urban and rural differences more appropriately for this type of peril. [2]

[3 Marks]

Solution 3:

Parameter Uncertainty refers to the uncertainty arising from the estimation of parameters used in the model.

Parameter uncertainty could arise from following sources:

- i. Data
 - a. Poor quality
 - b. Inconsistent
 - c. Incomplete
 - d. Non-existent
- ii. Changes in data format protocols
- iii. Insufficient data for certain perils
- iv. Changes in claims reserving philosophy

- v. Existence of
 - a. large claims
 - b. catastrophic claims
 - c. Latent claims
- vi. Changes in distribution channels leading to differential fixed and variable cost
- vii. Changes in claims inflation over the period

[4 Marks]**Solution 4:**

The company has credible historical data but many factors may cause the base experience to be different from that expected during the new rating period. Hence, we need to make suitable adjustments to both premium and claims data to be able to use it for future rating.

Following adjustments need to be made in premium/claims data:

- Adjust for any unusual heavy or light experience. Experience in any historical year can be unusually heavy/light may be due to catastrophe or climate change. It can be corrected by applying an adjustment factor to the affected year
- Adjustment for individual large or exceptional claims
- Adjusting for claims inflation; to project for future claim cost, need to allow for the effect of claims inflation
- Adjusting for changes in risk/cover provided – base period vs projected period
- Adjusting for changes in limits, excess and deductibles
- Changes in the method of distribution
- Changes in UW and Claim strategy
- Changes in reinsurance coverage
- Changes in perils covered
- Adjustment for external factors:
 - Legislative changes
 - Advancement in technology
 - Changes in construction of property in respect of material and technology

[4 Marks]**Solution 5:**

- Data Checks: Need to first establish that the data used for calculation are correct
 - Classifications of vehicles into high-end and low-end is correct
 - Whether the numerator and denominator of the claim frequency are consistent.
 - Whether treatment of nil claims is correct.
 - Data credible and reliable enough
- Time period of Data: The period of experience used may have a large effect on the figure
 - Time period might not correspond with that used for pricing (e.g. too old to be relevant or too recent to have shown in pricing yet)
 - Time period may be very short and statistically unreliable.
 - The periods may not be consistent for both sets of policies
- There is more to pricing than just the claim frequency. The cost of claims depends on claim amounts and not just claims frequency

- Need to check the claims amount and hence the loss ratio
- The office premium charged to the customer may differ from claims cost across levels of a rating factor if the company does not use constant expense loading across the book
- The behaviour of the policyholders with high-end cars shall be different from policyholders with low-end cars as high-end cars are generally well maintained and even a small accident (involving just a scratch or small dent) shall convert to a claim which might not be the case with low-end cars.
- There might be additional covers offered to high-end vehicle resulting in higher frequency but still having a low claim size.
- The CEO appears to be quoting a one-way analysis from the data and comparing with the output of GLM

[5 Marks]

Solution 6:

The goal of the analysis is to determine whether there is any expense cross-subsidy exist or not between two businesses.

For this, we can do detailed expense analysis to first determine, which expense arises as a result of writing each of the products, so that cost of writing each product can be determined.

- This cost will then be compared to the current expense loading for each product to see if there is indeed an expense cross-subsidy.
- The expense loading can be calculated by taking the gross premium and subtracting the expected claims cost and other loadings (e.g. capital loading and a negative loading for investment income).
- Expenses can be allocated between two products basis direct or indirect expenses and also fixed or variable expenses.

Other Analysis

The premium rates may be higher due to reasons other than expense cross-subsidy. We need to check for the same.

- The company's claim performance may be deteriorating resulting into higher premium.
- There may be an error in the pricing model used by the company which might be calculating wrong premium rates
- The company might be using distribution channels having higher acquisition cost
- There might be new players in the market who are charging lower premium to gain market share
- The drop in business may be due to falling service standards of the company

[6 Marks]

Solution 7:

i)

- An hour of sleep is well defined and objective, however it would be difficult to obtain and verify.
 - Obtaining data: hours slept per night is not often recorded by individuals, let alone reported.
 - Verifying data: You would have to rely on self-reporting which would be subject to unintentional mis-measurement or would encourage moral hazard and people would lie about it. Once they report it, it would be very difficult to verify.
- It's an ambiguous measure- hours in bed? Hours with eyes closed? What if insured gets up to drink water?

- It has been determined that sleep is a predictive rating variable, so it must be statistically significant, but could lack some credibility for groups at the high and low ends and could be difficult to be homogenous if people get more or less sleep every night.
- Information about sleeping habits would be considered a personal and unnecessarily intrusive question. People may be reluctant to discuss such matters with an insurance company.
- This would seem to have a causal relationship to motor own damage losses. People would be able to understand that not sleeping can increase the risk of losses. So from this standpoint, it would seem ok to use this as a rating variable.
- People will be more accepting if they can control to some extent what level of rating variable they belong to. Because you can adjust the hours you sleep. Using this criterion, it may be more accepted.

[4]

ii) Number of hours driven as an exposure base

- Vehicle year is easy to determine as compared to hours driven. It would require expensive telematics to determine
- Hours driven may or may not be proportional to expected loss. Some areas may be having more traffic congestion than others. Hence, more number of hours driven does not necessarily translate to higher expected loss
- The current exposure base in vehicle years. Changing to different exposure base may alter the premium rates significantly
- Changing the exposure base can be very expensive and time consuming. We need to restate the entire historical data for any analysis and trending.
- There may not be any data captured for this variable and so, may not be possible to restate the historical data. There may not be any industry benchmarks available for this exposure base
- Hours driven may not be appropriate as this can be manipulated by drivers if self-reported.

[3]

[7 Marks]

Solution 8:

i)

- The scheme was under Insurance Model for initial 8 years and under Trust model for the latest 2 years. Hence, the data may differ in terms of availability, credibility and completeness for both the periods as the way the data being managed shall be different for insurers and the State.
- The systems used by the State may not be suitable to record insurance data.
- The State system may be having issues in terms of data input, data validation and data extraction.
- The definition of claim may differ between various insurers and also with the State
- The data shared by previous insurers may be incomplete or inaccurate
- Previous insurers may withhold some data
- There may be a change in the policy coverage over the years. If that is not known, it shall affect the data analysis
- Since, for the last two years, the scheme is managed by the State, there shall not be any premium outflow and hence will be difficult to calculate claims frequency in the absence of underlying exposure measure.

[4]

ii) Checks for data reasonableness

- Check the number of members covered against census data
- Cross check the premium amount with the state agency as well as with the financials of the respective insurer.
- Check the reasonableness of number of claims. The same can be checked against the data published by various government agencies on number of accidental deaths for all years
- Check the amount paid against the type of claim and also the overall claim amount. The same can also be verified with the state agency and also the financials of the respective insurer. [2]

iii)

- To the technical premium calculated, add margin for contingency
- Use more reinsurance thus reducing the retention and hence the risk
- Be conservative in the pricing approach such as conservative assumptions.
- Do the sensitivity check of the model by varying underlying assumptions.

[2]

[8 Marks]**Solution 9:****i) Pure Premium Calculation:**

Working in Rs.'000

$$\text{Expected Loss} = (0 \times 0.95 + 0.02 \times 0.03 + 0.015 \times 0.14 + 0.01 \times 0.57 + 0.005 \times 1)M \\ = 0.0134M$$

$$\text{Pure Premium} = 0.0134M/M$$

$$= 0.0134 \text{ per 1000 SI}$$

[2]

ii) If M = 3500000, then the claims distribution shall be

Claim Payment	Probability
0	95%
100,000	2%
500,000	1.50%
2,000,000	1%
3,500,000	0.50%

For a policyholder buying a coverage of Rs.10,00,000

$$\text{Expected loss} = 0 \times 0.95 + 100 \times 0.02 + 500 \times 0.015 + 1000 \times (1 - 0.95 - 0.02 - 0.015) \\ = 24.5$$

$$\text{Premium charged at fixed rate} = 0.01343 \times 1000 \\ = 13.43$$

So premium is not equitable, expected loss of 24.5 vs premium of 13.43

[4]

iii) Purchasing a lower coverage than the true value of the home leads to underinsurance. In this scenario, there can be a coinsurance clause in the policy that would reduce the indemnity payment by the

proportion of selected coverage out of the required coverage. This would reduce the loss ratio of underinsured homes to the same loss ratio as fully insured homes.

The company should begin initiatives to increase the sum insured through home inspections e.g. making the underinsured homes to purchase the right amount. This would increase the premium for underinsured homes and equalize loss ratios.

[4]

[10 Marks]**Solution 10:**

- i) The possible risks faced by the resolution professionals under this code are
- The operational creditors may not get their dues/ recoveries under the resolution plan as expected and hence may file litigation against the resolution professionals citing reasons like improper execution of the plan or some errors/miscalculations (actual or alleged negligence) during the evaluation of the resolution plan.
 - Similarly the financial creditors may also sue the resolution professional for similar reasons as stated above.
 - The bidders may put charges against the resolution professional on grounds of not following rules/regulations laid down in the code during the bidding process. E.g. charge of showing biasness towards a particular bidder.
 - Risk of being kidnapped or coerced by the concerned parties to take a particular decision.
 - Risk of being framed for a bribery case.
 - Risk of reputation being tarnished for various reasons cited above.
 - Risk of sub-contracting going wrong. For example frauds committed by the sub-contractors. Here the sub-contracting means outsourcing some of the functions of the resolution professional to other third parties.
 - Risk of Data and confidentiality breaches due to reasons not under own control.
 - Risk of defence costs going much higher than expected during litigation.

[3]

- ii) The coverage's which can be sought by the resolution professionals broadly fall under and similar to two liability products (a) Errors or Omission/ Professional Indemnity (b) Directors and Officers Liability
The characteristics/features of these coverage's are

Benefits

- Defence costs towards defending against litigations for errors, negligence etc. while performing the duties of resolution professionals.
- Loss of income due to kidnapping and ransom, coercion, threat, etc.

Insured Perils

- Errors, Negligence, Delay, Kidnapping, Coercion, etc.

Basis of Cover

- Risk attaching basis with a discovery period clause.

- Losses occurring basis with a discovery period clause.

Measures of exposure to which premiums are related

- Total amount default loan outstanding
- Total Fees fixed for the resolution professionals

Claim Characteristics

- Low frequency
- High severity
- No reporting lag as such
- Settlement of claims can be delayed due to litigation procedures/process
- Chances of latent claims possible as this is an emerging area and it might be possible that the IBC code has some deficiencies.
- Low risk of fraudulent claims as the proceedings are very transparent.
- Volatility of claims may be high.
- Inflation may be mainly judiciary in nature.

Risk Factors and Rating Factors

Risk Factors

- Sector/ Industry to which the defaulting organization belongs to
- Level of defaulted debt in the overall industry to which the defaulting organisation belongs to
- Level of defaulted debt in the organisation
- Macro-economic factors like fiscal deficit of the country, GDP growth rate, Interest rate/ Inflation rate, exchange rate
- Level of Foreign participation allowed in order to enhance the competitiveness of the bidding process.
- Government rules and regulations applicable to the industry which would influence the success of a resolution process.
- Number of creditors involved ...the proportion of retail and institutional creditors including financial as well as operational creditors.
- Experience and skills of the resolution professionals
- Litigation system existing in the society to support quick litigation against the resolution professionals

Rating Factors

- Total defaulted debt of the defaulting organisation
- Total fees to be charged by the resolution professionals
- Years of experience relevant to the particular industry
- Number of such cases handled in the past
- Total defaulted debt of the entire sector to which the defaulting organisation belongs to
- Total number of creditors including all the financial, operational creditors, etc.

[8]

[11 Marks]

Solution 11:

i) Two broad categories are

○ Aggregate model

Here, the detailed information on the exposed risks (e.g. properties) is not known. Instead, aggregate exposures (e.g. sums insured) in an area are used in conjunction with industry average losses to estimate the likely losses. This works well as long as the actual risks insured are representative of industry averages, for example in terms of size and construction.

○ Detailed Model

Here, the individual insured risk information is used, and the likely loss for each insured risk is calculated, before summing to get aggregate losses. [1]

ii)

a) It is called Basis risk, more specifically it is called geographical basis risk. This risk pertains to the fact that the losses settled or received by the farmers may be way different than the actual losses incurred by the farmers because the overall geographical average loss may be different than that of an individual farmer's loss. [2]

b) It is the individual risk model which will help to eliminate this basis risk. The individual risk model will use the actual risk exposure of a farmer in terms of crop, acreage, cultivated land, etc and be able to assess the crop loss in respect of that acreage and cultivated land only. [2]

c) The challenges are:

- Very difficult to get all the individual farmland exposure information correctly. And the process of collection itself will be very costly given a vast geographical exposure.
- Extremely difficult to assess the individual farmland loss based on the individual exposure information as the administration cost is prohibitively high. India is a vast country with multiple crops, multiple weather patterns, multiple crop practices, etc. The cost of administration to manage this business over a large area at the individual farmland level is very high considering the high level of heterogeneity. Building the vulnerability module with this level of heterogeneity is extremely difficult and challenging.
- Administration at the individual farmland level is possible through technology to some extent but technology is also not reliable at times and may not be sophisticated yet to manage at this micro level.
- Moral hazard risk is very high i.e. chances of fraudulent claims will be very high if control checks are not implemented which anyway will be difficult to implement to manage the business at this micro level. These risks may not be modelled easily through any catastrophic model.
- Technology may be available but will have scalability issues and can be very costly also. This will enhance the cost of CAT modelling significantly.

- The hazard module and the event module may not be available at the localised level which will be required to assess the individual loss through pure modelling.
- The level of heterogeneity as mentioned above will also make it extremely difficult to build into the financial module as the terms and conditions may vary widely across a geographical area.
- Very few Cat Modellers will possess the required technology and expertise to build the individual risk model.
- If cropping patterns by the farmers change frequently, which is normally the case in India driven by so many factors like Minimum Support Price, Government subsidies, etc, then the individual risk model built on the past patterns will not hold credible for future prediction.
- Difficult to explain the variation of results in an individual risk model to a non-technical audience because of the nature of heterogeneity and complexity required in such models.

[4]

d) The challenges can be overcome by the following suggested ways:

- Using cutting edge recent technology like Artificial Intelligence and Machine Learning for prediction using multiple inputs. These technologies can really support the vulnerability module. Technology like Drones, SAR (Satellite aperture radar), satellite imaging, and remote sensing need to develop in a massive way.
- Technology has to be built upon the principle of scalability and adaptability at a cheaper cost. Government funding will be required at some of time.
- Data support systems need to develop quickly and in a large scale to support the physical hazard model used for crop loss assessment (vulnerability module) at the individual farmland level. The data support systems need to be developed with the help of local bodies and in a localised manner to tackle the heterogeneity issues.
- Promote research and development for building such individual risk models and spreading the awareness of benefits of these models in terms of reducing basis risk.
- Technology is the only solution to reduce frauds if implemented correctly.
- Replicate the models which are used abroad in global established markets.
- Develop and promote research through eminent institutions and academia in order to support development of hazard and event modules at localised level needed for such models.

[3]

[12 Marks]

Solution 12:

- i) The various sources of data error and their likely consequences on the financials of an insurance company are
- Wrong Claim number
Details of the claim could be allocated to the wrong record, and hence to the wrong claim risk group.
It can lead to incorrect pricing of the coverage's under a product which may lead to adverse loss experience of a portfolio.
 - Wrong Policy number

If the claim record picks up its risk details from the wrong policy record, these are likely to be wrong. Again, the claim details may be allocated to the wrong claim risk group, claim year and so on and therefore the incorrect pricing may lead to adverse loss experience of a portfolio.

- Wrong Risk Details

Actual Risk details are different at the time of claim which for some reasons are entered in the system post the claim event and before registration of claims.

Future claim assessment and prediction may go wrong and may lead to incorrect business plan.

The other error might be that the policy conditions have changed but the new policy conditions are still stored in the original rating group. This might again lead to incorrect future prediction and assessment of risk and eventually lead to incorrect business plan.

- Wrong Claim Date

A common cause of this is entering the date when the claim was notified, rather than the original incident date, or vice versa.

Reserving results may be different due to wrong allocation to different accident cohorts and erroneous development pattern. True financial position may be distorted because of this.

A wrong claim date could also mean that the claim will relate to the wrong risk details.

For liability claims and some property claims where a precise date of event cannot be determined, any wrong claim date may indicate alarming consequences.

- Wrong Payment Dates

Erroneous development patterns may distort the reserving results and hence true financial position may also be distorted accordingly.

- Wrong Claim Type

An analysis of different claim types for a particular risk group needs to be carried out as different claim types behave differently in terms of development and averages.

A change in the mix of types of claim also distorts the development patterns and average values.

Wrong claim types may lead to distorted pricing and reserving results which may adversely affect the loss experience.

[4]

- ii) The pricing of an adverse development cover is significantly dependent on the speed of claim settlements as this determines the level of investment income which may be earned on the reinsurance premiums received from the reinsured client.

The pricing of an adverse development factor is also dependent on the level of best estimate of the reserves held by the reinsured client.

Therefore the following sources of data error would significantly impact the pricing results

- **Wrong Claim Date**

Reserving results may be different due to wrong allocation to different accident cohorts and erroneous development pattern. True financial position may be distorted because of this.

A wrong claim date could also mean that the claim will relate to the wrong risk details.

For liability claims and some property claims where a precise date of event cannot be determined, any wrong claim date may indicate alarming consequences.

- **Wrong Payment Dates**

Erroneous development patterns may distort the reserving results and hence true financial position may also be distorted accordingly.

- **Wrong Claim Type**

A change in the mix of types of claim also distorts the development patterns and average values.

Wrong claim types may lead to distort the reserving results which may adversely affect the loss experience. [2]

iii)

a) The response to the student may be on the following lines

- If different claim number is allocated to a reopen claim then the number of claims will unnecessarily rise for a particular risk group and hence the frequency assessment will also go wrong.
- Similarly due to the reason above, the claim ticket size will also come down unnecessarily and hence the severity assessment will be distorted.
- Tail of claims reporting will look longer than what actual position is and may lead to higher Incurred But Not Reported Claims (IBNR) estimate than what should it should be
- Similarly the estimate towards Incurred But Not Enough Reported (IBNER) will decrease unnecessarily and be distorted.
- Depending on the reserving methodology, the reserving estimate may not depict the true estimate and be distorted instead.
- All the reasons mentioned above may finally lead to depiction of incorrect financial position of the company. [3]

- b) The correct practice is to allocate the original claim number to the re-open claim. Immediate challenges because of the change in the practice may be the following
- Cost towards the system change and implementation.
 - For a large organisation, this may require a cultural change with change in processes and systems.
 - Additional cost and effort towards training people explaining the change.

The consequences of the change may be the following

- Frequency will seem to come down drastically all of a sudden for certain risk groups and send false indication of the trends.
- Similarly the severity will seem to go up drastically all of a sudden for certain risk groups indicating false trends.
- The sudden change in the above trends will make it difficult it for making proper adjustments against these changes.

Depending on the particular way reserving is done , these sudden changes can distort the reserving results.

[3]

[12 Marks]

Solution 13:

i)

- a) The assumptions made under the Buhlmann-Straub Formula are
- There exists a latent parameter θ_i which can be characterised as the individual risk profile (here individual is referred here as i), which in itself is a random variable.

$E(X_{ik} | \theta_i) = \mu(\theta_i)$ (this is the underlying long run claims ratio of risk i)

$\text{Var}(X_{ik} | \theta_i) = \sigma^2(\theta_i) / V_{ik}$ (the variance of the observed claims ratio of risk i is inversely proportional to the volume measure of risk i).

- The i th risk is described by the pair, $(\theta_i, (X_{ik})_{k \geq 1})$ where $(X_{ik})_{k \geq 1}$ is the sequence of claims ratios observed for risk i in years k.
- The pairs $(\theta_i, (X_{ik})_{k \geq 1})$ are mutually independent.
- The θ_i are independent and identically distributed.
- Conditionally on θ_i , the X_{ik} 's are independent (but not necessarily identically distributed) [3]

- b) The Buhlmann-Straub model is similar to the Bayesian Credibility model. The similarity can be found in the form of the credibility formula which is $n / (n+k)$ in Bayesian credibility model. [2]

- ii) $C^{BE} = z_i X_i + (1 - z_i) \beta$,
 $z_i = V_i / [(V_i + \phi / \lambda)]$
 where $\beta = E(\mu(\theta_i))$,
 $\lambda = \text{Var}(\mu(\theta_i))$,
 $\phi = E(\sigma^2(\theta_i))$.

Here our target parameter is premium per family. Since we are considering only one risk i.e. a single family here, therefore the V_i is 1 here.

$$X_i = 2000$$

For Policyholders Category A

$$\mu(\theta_A) = \text{expected freq} * \text{expected severity} = 0.5714 * 7750 = 4428.571 \text{ per family}$$

For Policyholders Category B

$$\mu(\theta_B) = \text{expected freq} * \text{expected severity} = 1.00 * 5557.14 = 5557.14 \text{ per family}$$

From the given data, the chances of a family falling under each category is equal i.e. 50%.
(i.e. 50% = 7000/ 14000)

$$\beta = E(\mu(\theta_i)) = 50\% * \mu(\theta_A) + 50\% * \mu(\theta_B) = 4992.86$$

$$\begin{aligned} \lambda = \text{Var}(\mu(\theta_i)) &= E(\mu(\theta_i)^2) - E^2(\mu(\theta_i)) \\ &= 50\% * 4428.571^2 + 50\% * 5557.14^2 - 4992.86^2 \\ &= 318418.4 \end{aligned}$$

$$\text{Now, } \text{Var}(X) = E[\text{Var}(X) | \theta] + \text{Var}[E(X) | \theta]$$

$$= \phi + \lambda$$

$$\text{Since } \text{Var}(X) \text{ is given, therefore } \phi = \text{Var}(X) - \lambda = 2300000000 - 318418.4 = 229681582$$

$$\begin{aligned} \text{Hence } z_i &= V_i / [(V_i + \phi / \lambda)] = 1 / [1 + \phi / \lambda] = 1 / [1 + (229681582 / 318418.4)] \\ &= 1 / 722.32 = 0.14\% \end{aligned}$$

Therefore the suggested premium which should be charged to the family as per the Buhlmann Straub Model is $0.14\% * 2000 + 99.86\% * 4992.86 = 4988.67$ [7]

iii) The assumptions which may be violated under the specific problem

- The pairs $(\theta_i, (X_{ik})_{k \geq 1})$ are mutually independent.

The pairs may not be actually independent as a group of risks may be subjected to a particular peril because of their geographical presence, distribution channel and other demographics such as lifestyle, community cultures, etc. The observed losses from a group of risks may also have arisen because of a particular fraud impacting a large number of risks at the same time. The losses may also have arisen because of an epidemic which might be covered under the scheme.

- The θ_i are independent and identically distributed.

This independence assumption may be violated as a group of risks may belong to a particular tribe with similar lifestyle, working and living conditions.

The assumption of identical distribution may be violated as the inherent factors driving the claims from a group of risks may be varying and cannot be standardised through a single parameter or a distribution. If there are too many heterogeneous groups and communities with too much of diversity, this assumption may not hold.

- Conditionally on θ_i , the X_{ik} 's are independent (but not necessarily identically distributed)

Even with the conditional on θ_i , the X_{ik} 's may not be independent as still there will be some common drivers which will be influencing the losses in a particular direction for a group of risks. The inherent nature of such common drivers is such that they cannot be modelled with a specific distribution as such.

[3]

[15 Marks]
